

Getting A Girlfriend



Life at NITK is awesome for everyone but have you heard about failed_coder? He is a geek at NITK and his life is playing a game with him. He is trying hard to get a girlfriend but still he hasn't got. This time he has come up with an idea. He selects a group of girls and then from the group he purposes a girl. As he is very smart, he always chooses such a group that there is maximum hope. What is the maximum probability that he will get a girlfriend this time? Every girl has a status whether she has a partner or not. If a girl has a partner then she must reject failed_coder. Also, as there are very less girls at NITK so every girl has got some proposals already. If a girl has some proposals then there is same probability for every proposal to be accepted.

Input Format

The first line of input contains an integer n denoting the number of girls.

The second line of input contains n space separated integers $A_1, A_2, \dots, A_n$. Each $A_i=0$ or 1 . 0 shows that i th girl has no partner and 1 shows that i th girl has a partner.

The third line of input contains n space separated integers $B_1, B_2, \dots, B_n$. B_i shows the number of proposals i th girl has got before failed_coder's proposal.

Constraints

$1 \leq n \leq 5000$.

$A_i = 0$ or 1 for every $1 \leq i \leq n$.

$0 \leq B_i \leq 10^9$ for every $1 \leq i \leq n$.

Output Format

Print the maximum probability that failed_coder will get a girlfriend.

The output should not have absolute error of more than 10^{-6} .

Sample Input 0

```
2
1 1
2 3
```

Sample Output 0

```
0
```

Explanation 0

Every girl already has a partner so they will reject him.

Sample Input 1

```
2
0 0
1 1
```

Sample Output 1

```
0.5
```

Explanation 1

Every girl has 2 proposals(1 already + 1 failed_coder's) so there is 50% chance.

